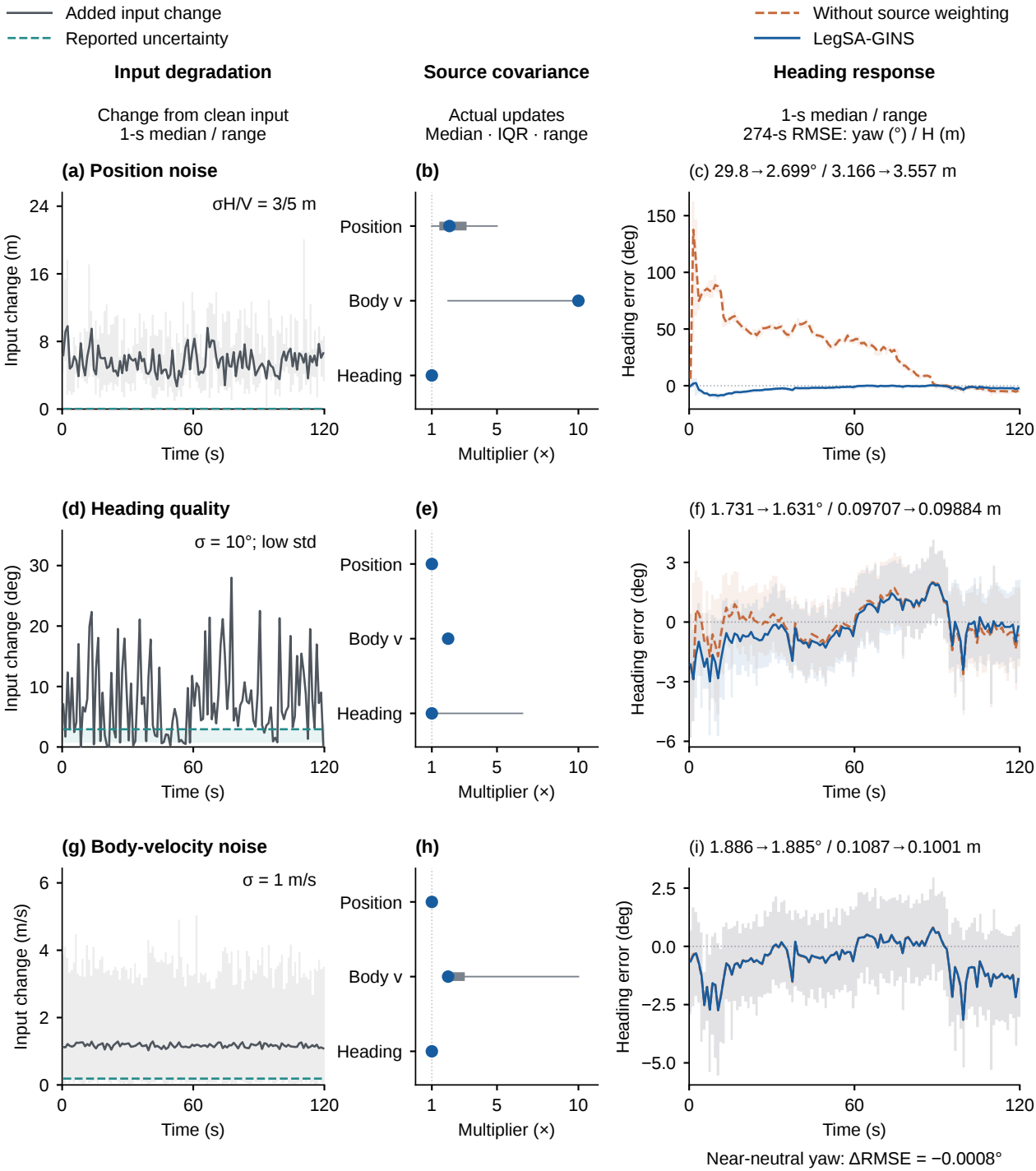

Added input change
Reported uncertainty
Without source weighting
LegSA-GINS
Input degradation
Source covariance
Heading response
Change from clean input
1-s median / range
Actual updates
Median · IQR · range
1-s median / range
274-s RMSE: yaw (°) / H (m)
(a) Position noise
σH/V = 3/5 m
Input change (m)
Time (s)
(b)
Position
Body v
Heading
Multiplier (×)
(c) 29.8→2.699° / 3.166→3.557 m
Heading error (deg)
Time (s)
(d) Heading quality
σ = 10°; low std
Input change (deg)
Time (s)
(e)
Position
Body v
Heading
Multiplier (×)
(f) 1.731→1.631° / 0.09707→0.09884 m
Heading error (deg)
Time (s)
(g) Body-velocity noise
σ = 1 m/s
Input change (m/s)
Time (s)
(h)
Position
Body v
Heading
Multiplier (×)
(i) 1.886→1.885° / 0.1087→0.1001 m
Heading error (deg)
Time (s)
Near-neutral yaw: ΔRMSE = −0.0008°